MediPredict: Multi-Modal Disease Prediction Using Clinical and Laboratory Data with Explainable Artificial Intelligence

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ABSTRACT

Early screening of chronic diseases can reduce avoidable complications, treatment delay, and clinical workload. However, many machine-learning-based diagnostic systems train one model for one disease and often ignore the complementary value of clinical measurements and laboratory markers. This paper presents MediPredict, a unified machine learning framework for predicting diabetes, heart disease, chronic kidney disease, and liver disease from publicly available medical datasets. The framework cleans heterogeneous data, separates clinical and laboratory feature groups, applies ANOVA-based feature selection, handles class imbalance using SMOTE inside the training pipeline, and compares six supervised classifiers under a stratified cross-validation protocol. The evaluated models include logistic regression, decision tree, random forest, support vector machine, gradient boosting, and XGBoost. MediPredict also integrates explainable artificial intelligence through SHAP and LIME to expose global feature importance and local patient-level reasoning. Experimental results show strong performance for heart disease and kidney disease, while diabetes and liver disease remain more challenging and clinically realistic classification tasks. Ablation results indicate that combined clinical and laboratory features improve F1-score and ROC-AUC over single-modality inputs for diabetes and liver disease. The proposed system supports low-cost, interpretable, multi-disease risk screening and provides a foundation for deployment through a Streamlit-based clinical decision-support interface.

Index Terms--Chronic disease prediction, explainable artificial intelligence, SHAP, LIME, SMOTE, machine learning, clinical decision support, multi-modal data fusion.

I. INTRODUCTION

Noncommunicable diseases impose a major burden on global health systems. Cardiovascular diseases remain the leading cause of death globally, and diabetes, chronic kidney disease, and liver disorders contribute substantially to long-term morbidity. In routine clinical practice, physicians rarely rely on a single measurement. They interpret demographic history, symptoms, bedside measurements, and laboratory markers together. A computational disease prediction system should therefore use the same principle: it should combine clinical and laboratory evidence, provide calibrated risk estimates, and explain the factors that drive each prediction.

Existing machine learning studies commonly focus on a single disease dataset and report accuracy from one or two classifiers. This design limits cross-disease comparison and makes deployment difficult in resource-limited settings where a single software platform must support multiple screening tasks. Furthermore, high accuracy alone does not establish clinical trust. A doctor must know why a model predicted risk before using the output in triage or counseling. MediPredict addresses this gap by combining four disease domains, six supervised algorithms, ablation analysis, statistical testing, and explainable AI in one reproducible workflow.

This paper makes four main contributions. First, it proposes a unified prediction pipeline for diabetes, heart disease, chronic kidney disease, and liver disease. Second, it evaluates classical and ensemble machine learning models using stratified cross-validation, SMOTE, and weighted performance metrics. Third, it compares clinical-only, laboratory-only, and combined feature settings to quantify the value of multi-modal fusion. Fourth, it uses SHAP and LIME to translate model behavior into interpretable feature contributions for global and patient-level explanations.

II. RELATED WORK

Machine learning has been widely applied to chronic disease prediction. Diabetes studies often use the Pima Indians Diabetes dataset and compare logistic regression, support vector machines, random forests, boosting algorithms, and neural networks. These studies report useful baseline performance but frequently limit evaluation to one disease and one tabular dataset. Heart disease prediction work has similarly used UCI-derived datasets with ensemble methods and, more recently, explainability tools such as SHAP and LIME. Chronic kidney disease studies have reported high classification accuracy on the UCI CKD dataset, but several approaches remain vulnerable to leakage if identifier columns or post-diagnosis fields enter the training process. Liver disease prediction studies using the Indian Liver Patient Dataset have explored preprocessing, resampling, feature selection, and ensemble learning because the dataset is imbalanced and clinically noisy.

Explainable AI methods have become important in healthcare because they expose the relationship between features and model outputs. SHAP estimates feature contributions using Shapley-value principles and supports both global and local interpretation. LIME explains an individual prediction by training a local surrogate model around one patient instance. Prior studies often apply either SHAP or LIME to one disease. MediPredict extends this direction by applying both methods across four disease domains and by connecting interpretability with multi-modal feature analysis.

III. PROPOSED METHODOLOGY

Fig. 1 shows the overall MediPredict workflow. The system loads raw disease datasets, standardizes heterogeneous feature types, removes invalid values, encodes categorical variables, applies feature selection, and trains disease-specific models. The final prediction interface accepts patient inputs, returns disease risk, stores prediction history, and presents explanatory evidence through SHAP and LIME visualizations.

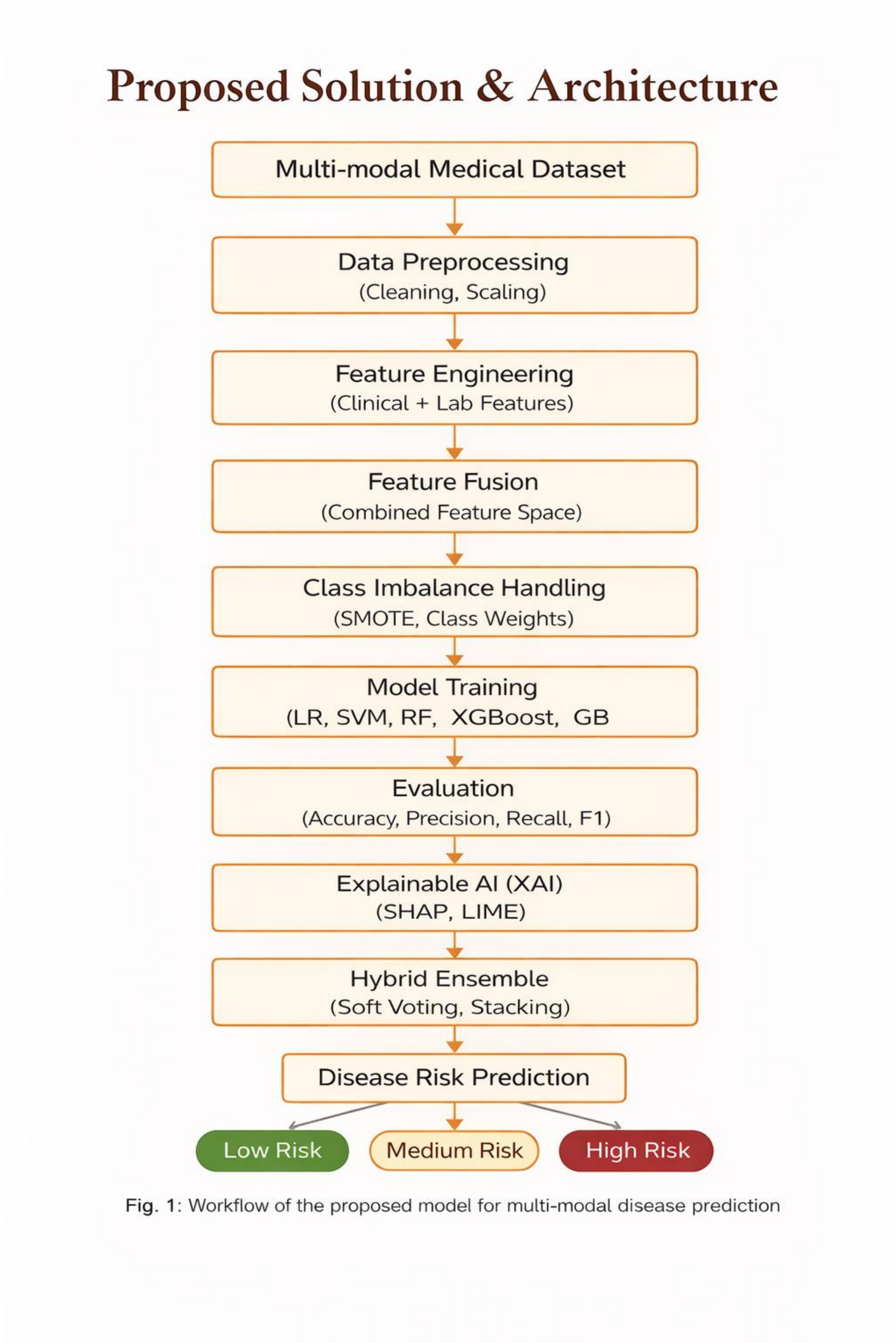

Fig. 1. Proposed solution and system architecture of the MediPredict framework.

A. Dataset Description

The study uses four public tabular datasets. The diabetes module uses the Pima Indians Diabetes dataset with 768 records and eight predictors. The heart disease module uses a UCI/Kaggle heart disease dataset with 1,025 records and clinical/laboratory predictors. The kidney module uses the UCI chronic kidney disease dataset with 400 records and mixed numerical-categorical attributes. The liver module uses the Indian Liver Patient Dataset with biochemical markers collected from patients in India.

TABLE I. DATASETS USED IN THE MEDIPREDICT STUDY

Disease	Dataset	Records	Target	Main Feature Types
Diabetes	Pima Indians Diabetes	768	Outcome	Glucose, BMI, age, insulin, blood pressure
Heart	UCI/Kaggle Heart Disease	1,025	target	Chest pain, cholesterol, oldpeak, thalach, ca, thal
Kidney	UCI CKD	400	classification	Blood pressure, albumin, sugar, hemoglobin, creatinine
Liver	ILPD	583/584	Dataset	Bilirubin, liver enzymes, proteins, albumin, age, gender

B. Preprocessing and Feature Selection

The preprocessing stage treats disease-specific data quality issues before model training. For diabetes, physiologically invalid zero values in glucose, blood pressure, skin thickness, insulin, and BMI are replaced with missing values and imputed using the median. For heart disease, missing numerical values are imputed using the median. For kidney disease, question marks and inconsistent string values are normalized, clinical categories such as yes/no, present/not present, and normal/abnormal are encoded numerically, and missing numerical values are imputed using the mean. For liver disease, gender is encoded as a binary variable, missing values are imputed using the median, and the target is mapped to binary labels.

After cleaning, the framework applies ANOVA F-test feature selection for each disease. This step ranks predictors by their class-discriminative power and reduces weak predictors before model training. Fig. 2 presents representative ANOVA feature-importance plots for heart, kidney, and liver disease.

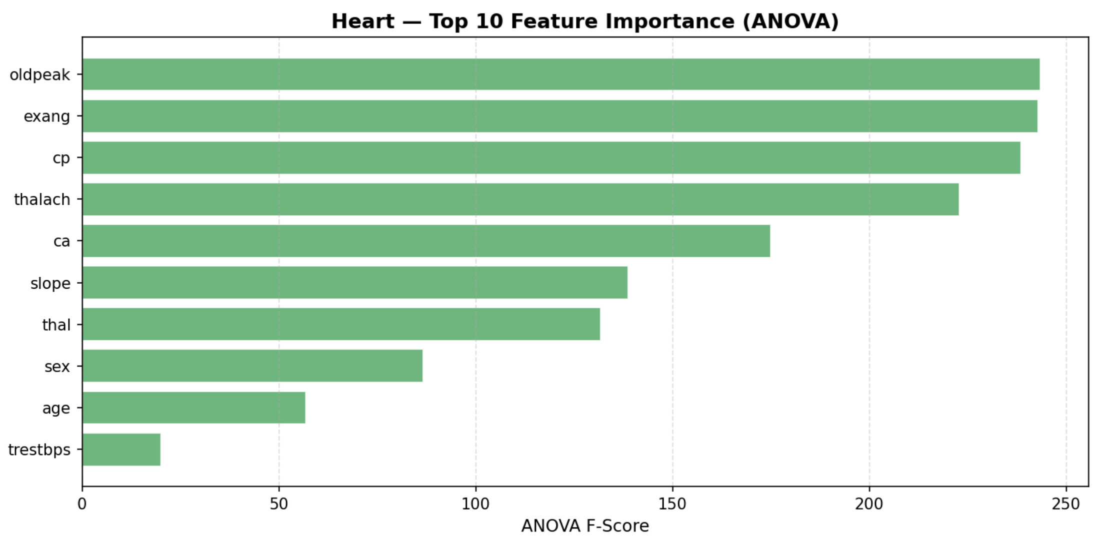

Fig. 2(a). ANOVA feature ranking for heart disease.

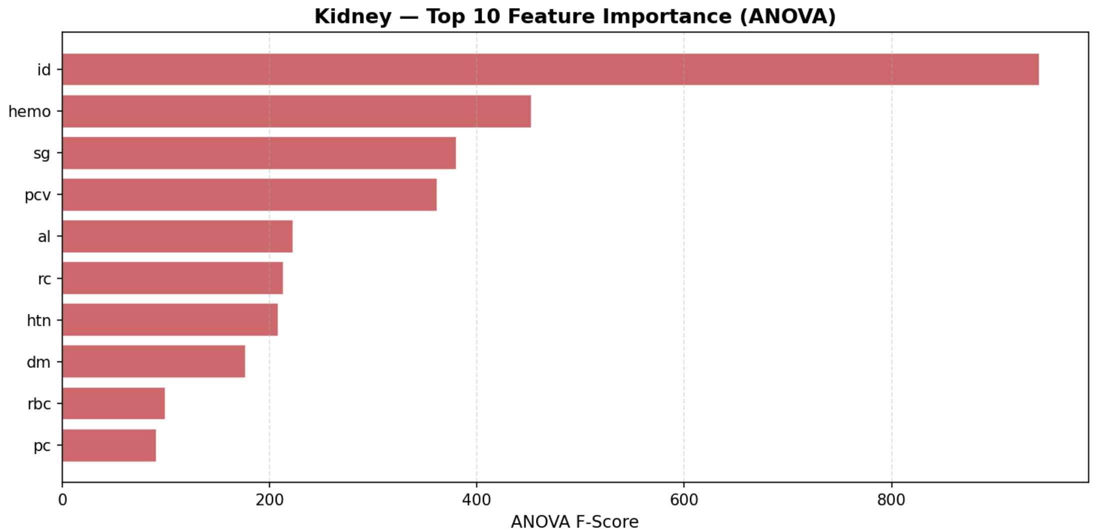

Fig. 2(b). ANOVA feature ranking for chronic kidney disease.

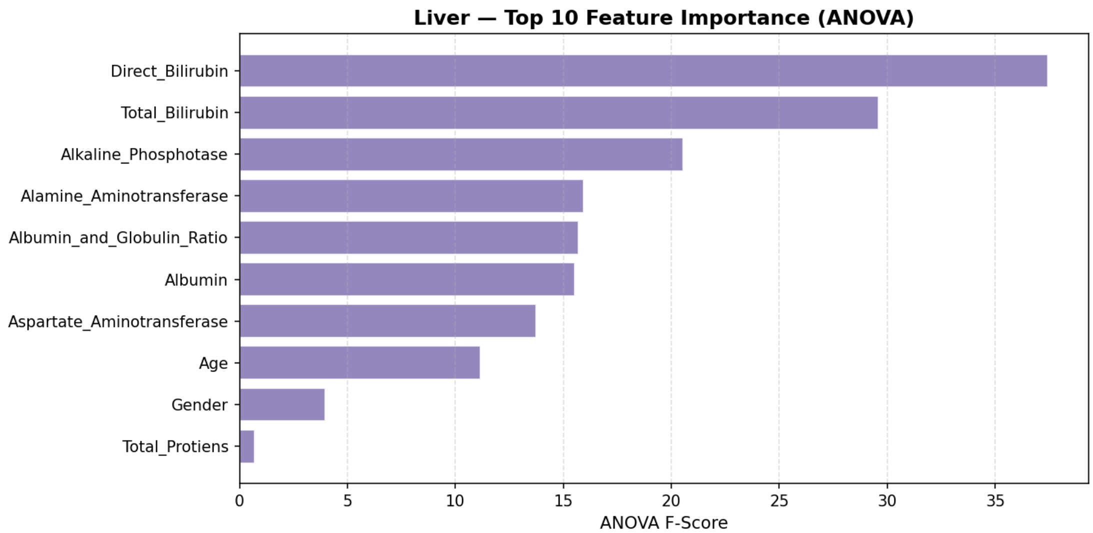

Fig. 2(c). ANOVA feature ranking for liver disease.

C. Model Training Pipeline

Each disease-specific model follows the same training pipeline: SMOTE, standard scaling, and classifier training. The use of SMOTE inside the imbalanced-learn pipeline prevents data leakage because synthetic samples are generated only within the training folds. Hyperparameters are tuned using randomized search with inner five-fold stratified cross-validation. The final evaluation uses outer ten-fold stratified cross-validation. The framework reports accuracy, weighted precision, weighted recall, weighted F1-score, and ROC-AUC. Weighted metrics are used because the datasets contain imbalanced class distributions.

The evaluated algorithms include logistic regression, decision tree, random forest, support vector machine, gradient boosting, and XGBoost. The best model for each disease is selected primarily by weighted F1-score because this metric balances false positives and false negatives under class imbalance.

IV. EXPERIMENTAL RESULTS

Tables II-V summarize the ten-fold cross-validation results for all four disease modules. The results are copied from the supplied training outputs. Heart disease and kidney disease show near-perfect scores for several models, while diabetes and liver disease show moderate performance. This difference suggests that the latter two datasets contain more overlapping class boundaries and noisier predictors.

TABLE II. DIABETES CLASSIFICATION RESULTS

Algorithm	Accuracy	F1-score	ROC-AUC
Logistic Regression	0.7524	0.7547	0.8354
Decision Tree	0.7173	0.7198	0.7447

Random Forest	0.7576	0.7599	0.8312
SVM	0.7537	0.7567	0.8341
Gradient Boosting	0.7602	0.7642	0.8349
XGBoost	0.7603	0.7634	0.8297

TABLE III. HEART DISEASE CLASSIFICATION RESULTS

Algorithm	Accuracy	F1-score	ROC-AUC
Logistic Regression	0.8400	0.8391	0.9190
Decision Tree	0.9786	0.9786	0.9991
Random Forest	1.0000	1.0000	1.0000
SVM	0.9971	0.9971	0.9978
Gradient Boosting	1.0000	1.0000	1.0000
XGBoost	0.9912	0.9912	0.9997

TABLE IV. CHRONIC KIDNEY DISEASE CLASSIFICATION RESULTS

Algorithm	Accuracy	F1-score	ROC-AUC
Logistic Regression	0.9975	0.9975	1.0000
Decision Tree	1.0000	1.0000	1.0000
Random Forest	1.0000	1.0000	1.0000
SVM	1.0000	1.0000	1.0000
Gradient Boosting	1.0000	1.0000	1.0000
XGBoost	0.9975	0.9975	1.0000

TABLE V. LIVER DISEASE CLASSIFICATION RESULTS

Algorithm	Accuracy	F1-score	ROC-AUC
Logistic Regression	0.6428	0.6583	0.7403
Decision Tree	0.6378	0.6451	0.6053
Random Forest	0.7271	0.7295	0.7755
SVM	0.6377	0.6528	0.7307
Gradient Boosting	0.6824	0.6912	0.7387
XGBoost	0.6979	0.7004	0.7391

A. Ensemble and Ablation Analysis

A soft-voting ensemble was evaluated using random forest, gradient boosting, and XGBoost. The ensemble reached F1-scores of 0.7567 for diabetes, 1.0000 for heart disease, 0.9975 for kidney disease, and 0.7204 for liver disease. These results show that ensembling preserves strong performance on separable datasets but does not automatically solve difficult class boundaries in diabetes and liver disease.

TABLE VI. SOFT-VOTING ENSEMBLE RESULTS

Disease	Accuracy	F1-score	ROC-AUC
Diabetes	0.7551	0.7567	0.8219
Heart	1.0000	1.0000	1.0000
Kidney	0.9975	0.9975	1.0000
Liver	0.7219	0.7204	0.7725

The ablation study compares clinical-only, laboratory-only, and combined feature settings using random forest. Combined features improve the F1-score and ROC-AUC for diabetes and liver disease, which supports the central hypothesis that clinical and laboratory variables provide complementary evidence. Heart and kidney disease remain near-perfect under multiple settings, so those results require cautious interpretation during final validation.

TABLE VII. ABLATION RESULTS FOR CLINICAL, LABORATORY, AND COMBINED FEATURES

Disease	Clinical F1/AUC	Lab F1/AUC	Combined F1/AUC
Diabetes	0.664 / 0.707	0.731 / 0.787	0.754 / 0.820
Heart	0.991 / 0.999	1.000 / 1.000	1.000 / 1.000
Kidney	0.980 / 0.999	0.967 / 0.992	0.992 / 1.000

Liver	0.666 / 0.690	0.680 / 0.714	0.714 / 0.777

V. EXPLAINABILITY AND INTERPRETABILITY

MediPredict uses SHAP for global feature attribution and LIME for local patient-level explanations. SHAP identifies the predictors that contribute most strongly to model output across the test set. LIME explains one patient at a time by approximating the model locally and showing which features increase or decrease disease risk.

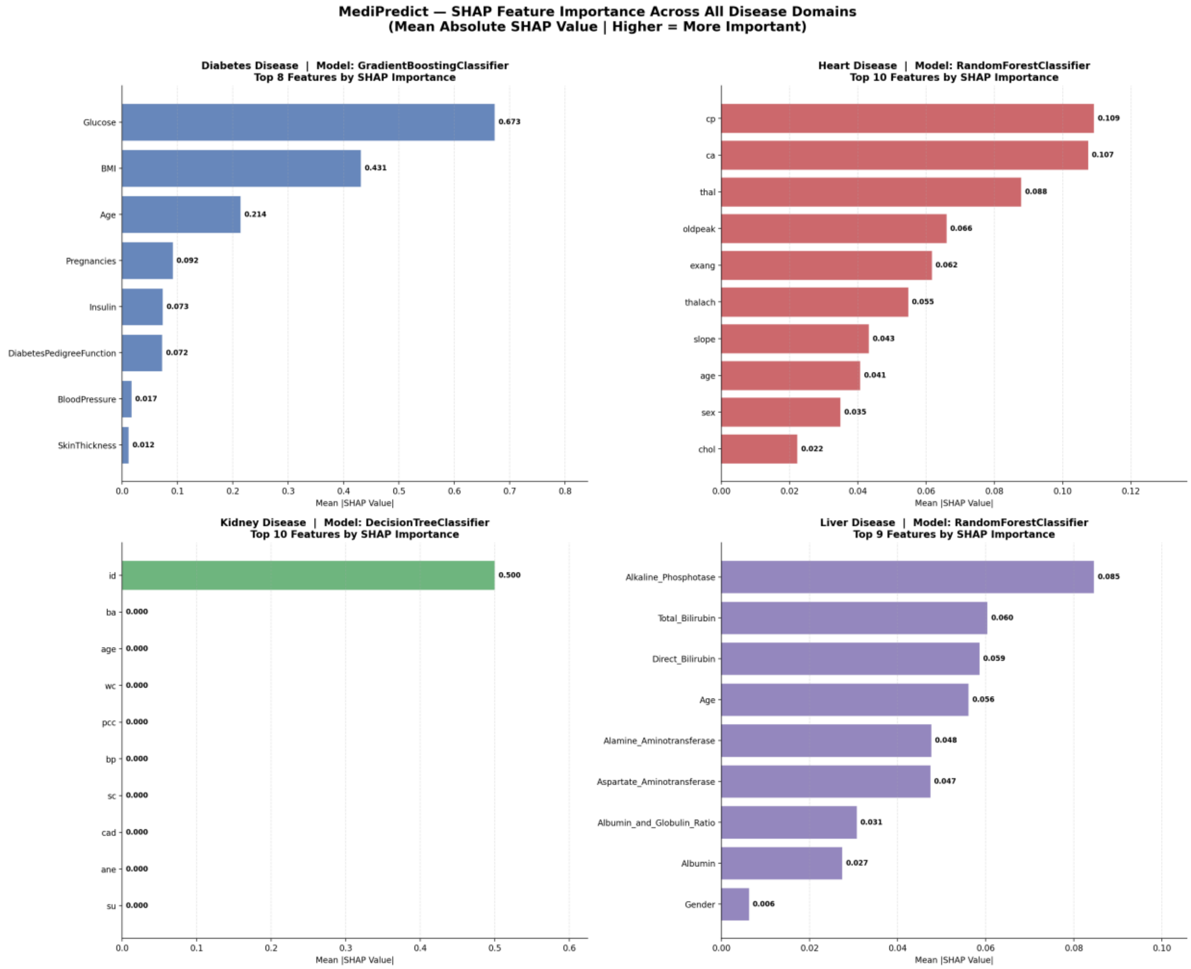

Fig. 3. SHAP feature-importance comparison across all disease domains.

The SHAP comparison shows that glucose, BMI, and age dominate diabetes prediction. In the heart disease model, chest pain type, number of major vessels, thalassemia status, oldpeak, and exercise-induced angina contribute most strongly. For liver disease, alkaline phosphatase, total bilirubin, direct bilirubin, age, and alanine aminotransferase have the highest mean absolute SHAP values. The kidney result identifies the identifier field as a dominant feature, which should be treated as a preliminary artifact and removed during the next training iteration.

TABLE VIII. TOP SHAP FEATURES FROM THE XAI SUMMARY

Disease	Top SHAP Features	Best Model Used for XAI
Diabetes	Glucose, BMI, Age, Pregnancies, Insulin	Gradient Boosting
Heart	cp, ca, thal, oldpeak, exang	Random Forest
Kidney	id, ba, age, wc, pcc	Decision Tree (preliminary)
Liver	Alkaline phosphatase, Total bilirubin, Direct bilirubin, Age, ALT	Random Forest

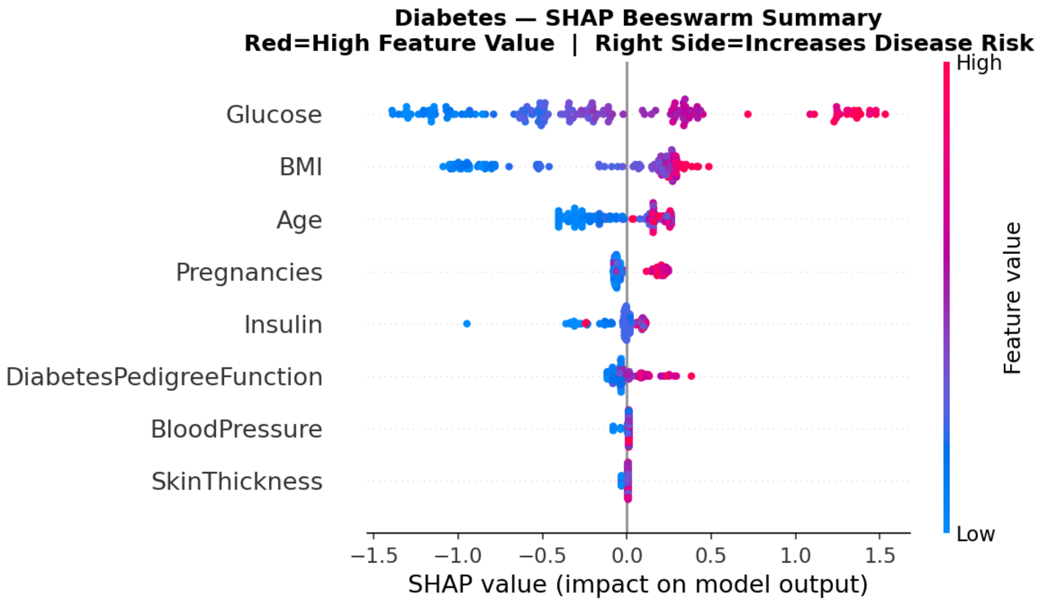

Fig. 4. SHAP beeswarm summary for diabetes prediction.

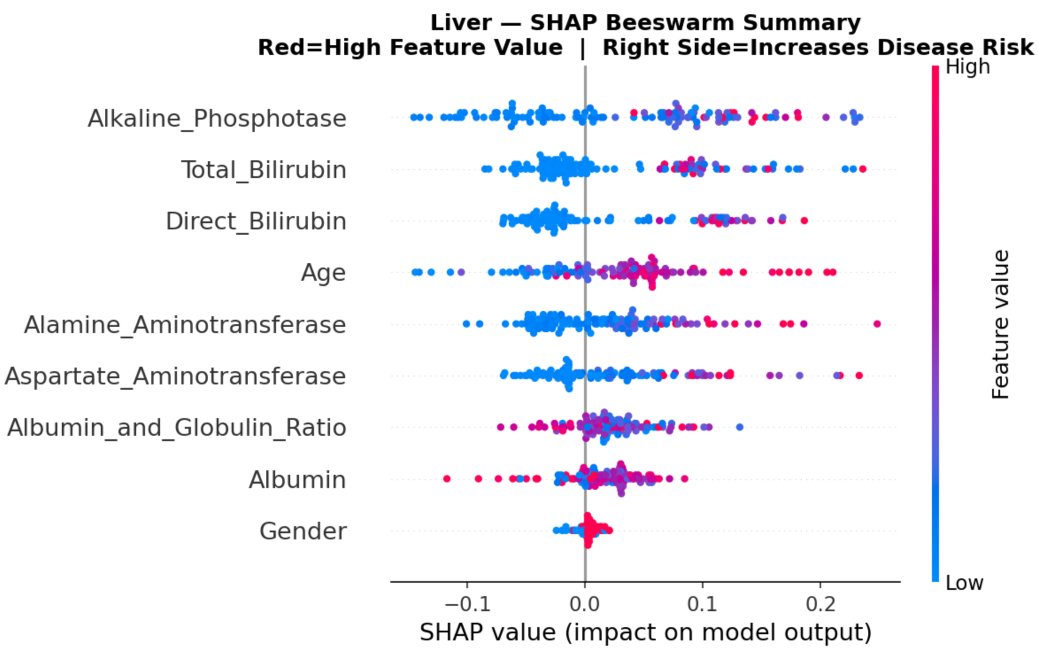

Fig. 5. SHAP beeswarm summary for liver disease prediction.

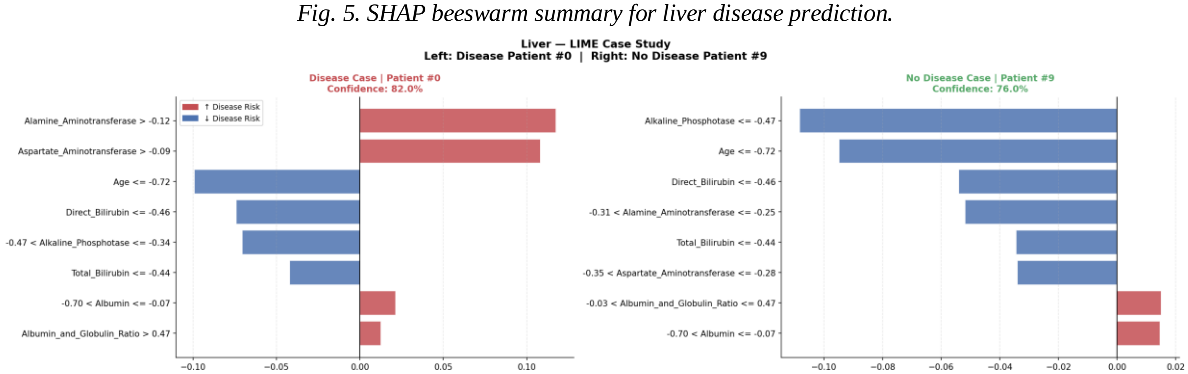

Fig. 6. LIME case study for liver disease showing disease and no-disease patient explanations.

The diabetes beeswarm plot shows that high glucose and high BMI values push predictions toward disease risk, while lower values push predictions toward the no-disease class. The liver beeswarm plot shows that bilirubin markers and liver enzymes drive large changes in model output. The LIME case study complements SHAP by showing how individual marker ranges push a single patient toward disease or no-disease classification.

VI. DISCUSSION

The results support the usefulness of a unified chronic disease screening framework, but they also reveal the limitations of benchmark datasets. Diabetes and liver disease results are moderate and therefore more realistic for clinical screening. Heart disease results are extremely high, which may reflect dataset separability, duplication, or strong feature-target structure. Kidney disease results require additional validation because the current feature set includes an identifier column that can behave as a leakage-prone predictor. The next version of this work should remove non-clinical identifiers, retrain the kidney model, and regenerate SHAP values before final publication.

The ablation study provides a clinically meaningful finding. Combined clinical and laboratory features improve diabetes and liver prediction over clinical-only inputs. This supports the multi-modal design of MediPredict. In practice, a screening system should avoid relying only on symptoms or only on laboratory values when both are available. The explainability module also improves practical trust because it converts model output into clinical reasoning cues, such as glucose and BMI for diabetes or bilirubin and liver enzymes for liver disease.

VII. LIMITATIONS

This study has several limitations. First, the datasets are public benchmark datasets and may not represent the distribution of patients in a real hospital. Second, the current system uses tabular clinical and laboratory features only; it does not yet include imaging, prescriptions, physician notes, or longitudinal follow-up. Third, some models achieved near-perfect scores on heart and kidney datasets, which requires duplicate checking, leakage analysis, and external validation. Fourth, the kidney model must be retrained after removing the identifier column. Fifth, the current Streamlit interface supports decision support and screening, not independent diagnosis.

VIII. CONCLUSION AND FUTURE WORK

This paper presented MediPredict, a multi-disease prediction framework that combines clinical and laboratory data with explainable artificial intelligence. The system standardizes preprocessing across four chronic disease domains, compares six supervised classifiers, applies SMOTE inside cross-validation, evaluates ablation settings, and uses SHAP and LIME to explain global and local model behavior. The results show that combined clinical and laboratory features improve performance for diabetes and liver disease, while explainability highlights clinically relevant predictors such as glucose, BMI, bilirubin, and liver enzymes. Future work will remove non-clinical leakage-prone fields, retrain the kidney model, add LightGBM and neural network baselines if implemented, validate the system on external hospital data, and deploy the Streamlit application with calibrated risk categories and secure prediction-history storage.

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